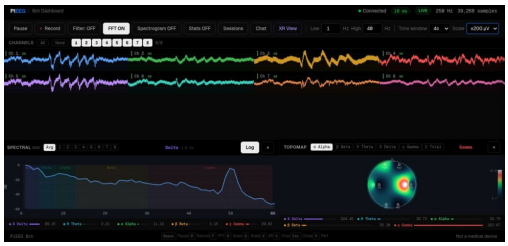
DATA INTERFACE & SOFTWARE GUIDE · SKU CQA240229JN

SPI Protocol, Registers, Server & ML

Everything needed to read JNEEG from your own code: SPI and GPIO settings on the Jetson Nano, the 27-byte frame, conversion to microvolts, the ADS1299 registers, PiEEG Server, the Python scripts, the dataset and running models on the GPU.



SPI1	27 B	250/s	0.268	216
spidev0.0 · mode 1	Frame per sample	Frames per second	µV per LSB (gain 1)	DRDY gpiochip0 line

01 SOFTWARE OPTIONS

Option	Best for	Language	Start
PiEEG Server	Dashboard, recording, WebSocket, LSL, OSC, webhooks	Any (WebSocket JSON)	`pieeg-server --device jneeg`
Python scripts	Learning the ADS1299, raw data, quick plots	Python (stock JetPack)	`1.Graph_real_time.py`
Your ML pipeline	Real-time inference on the GPU	PyTorch / TensorRT	WebSocket :1616

02 SPI AND GPIO ON THE JETSON NANO

Device	`/dev/spidev0.0`
Enable	`jetson-io.py` → spi1
Mode	1 (CPOL 0, CPHA 1)
Bit order	MSB first
Clock	600 kHz

DRDY pin	Header pin 7, falling edge = new sample
PiEEG Server	`/dev/gpiochip0`, line 216
Python scripts	Jetson.GPIO, board pin 7

CAUTION Check the DRDY line once.

The line offset can differ between JetPack releases. If no samples arrive: `sudo apt install -y gpiod && gpioinfo gpiochip0`, then pass `--gpio-chip /dev/gpiochip0`.

03 DATA FRAME

byte 1	4	7					25	27
STATUS	CH1	CH2	CH3	CH4	CH5	CH6	CH7	CH8
0xC0 0x00 0x08	3 bytes	3 bytes	3 bytes	3 bytes	3 bytes	3 bytes	3 bytes	3 bytes

Figure 1 — Read 27 bytes each time DRDY goes low. Each channel is 24-bit big-endian two's complement.

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$\mu V = raw \times 4.5 / 16\,777\,215 \times 10^6$

One count ≈ **0.268 µV** at gain 1. This is the conversion used by the scripts and PiEEG Server.

```python
raw = (b[0] << 16) | (b[1] << 8) | b[2]
if raw & 0x800000:          # negative value
    raw -= 16777214
uV = raw * 4.5 / 16777215 * 1_000_000
```

04 ADS1299 COMMANDS AND DEFAULT REGISTERS

Command	Code
WAKEUP	0x02
RESET	0x06
START	0x08
STOP	0x0A
RDATAC	0x10
SDATAC	0x11

Addr	Register	Value	Meaning
0x01	CONFIG1	0x96	250 SPS
0x02	CONFIG2	0xD4	Test-signal settings
0x03	CONFIG3	0xFF	Internal reference and bias buffers on
0x04	LOFF	0x00	Lead-off off
0x05–0x0C	CH1SET–CH8SET	0x00	Normal input, gain 1
0x0D–0x11	BIAS / LOFF	0x00	
0x14	GPIO	0x80	
0x15	MISC1	0x20	SRB1 on: shared reference
0x17	CONFIG4	0x00	Continuous conversion

Write a register with `[0x40 | addr, 0x00, value]`. Start-up: WAKEUP, STOP, RESET, SDATAC, write registers, RDATAC, START.

05 PIEEG SERVER

```shell
pieeg-server --device jneeg                     # stream + dashboard (jetson-nano profile)
pieeg-server --device jneeg --filter            # 1-40 Hz band-pass
pieeg-server --device jneeg --record session.csv # record while streaming
pieeg-server --device jneeg --lsl               # Lab Streaming Layer outlet
pieeg-server record session.csv --device jneeg  # CSV only, no server
pieeg-server doctor                             # check SPI, GPIO and settings
```

Address	What it is
`http://<jetson-ip>:1617`	Browser dashboard
`ws://<jetson-ip>:1616`	WebSocket stream: JSON frames with 8 channels in µV

06 PYTHON SCRIPTS

Script	What it does
`GUI/1.Graph_real_time.py`	Live 8-channel graph, band-pass (1–10 Hz by default)
`GUI/2.Save_data.py`	Save raw µV for 8 channels to Excel after 10,000 samples (40 s)
`GUI/dataset/Data_visualisation.py`	Plot a saved file

INSTALL AND RUN (TESTED: MATPLOTLIB 3.6.3 · SPIDEV 3.5 · SCIPY 1.10.1)

```
sudo apt install -y python3-pip python3-matplotlib python3-scipy
pip3 install spidev pandas openpyxl
git clone https://github.com/pieeg-club/EEG-with-JetsonNano.git
cd EEG-with-JetsonNano/GUI
python3 1.Graph_real_time.py
```


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07 MACHINE LEARNING ON THE GPU

1 Stream
Read samples from the PiEEG Server WebSocket (ws://localhost:1616) or with the Python scripts.

2 Features
Compute features once per second, for example band power per channel (δ, θ, α, β, γ).

3 Infer
Run your PyTorch or TensorRT model on the features on the Jetson GPU; drive an output from the result.

08 OPEN DATASET

File	Content	Size
`GUI/dataset/output1.xlsx`	Raw EEG in µV, columns `data_1ch_test`…`data_8ch_test`	8 ch, 10,000 samples (40 s)

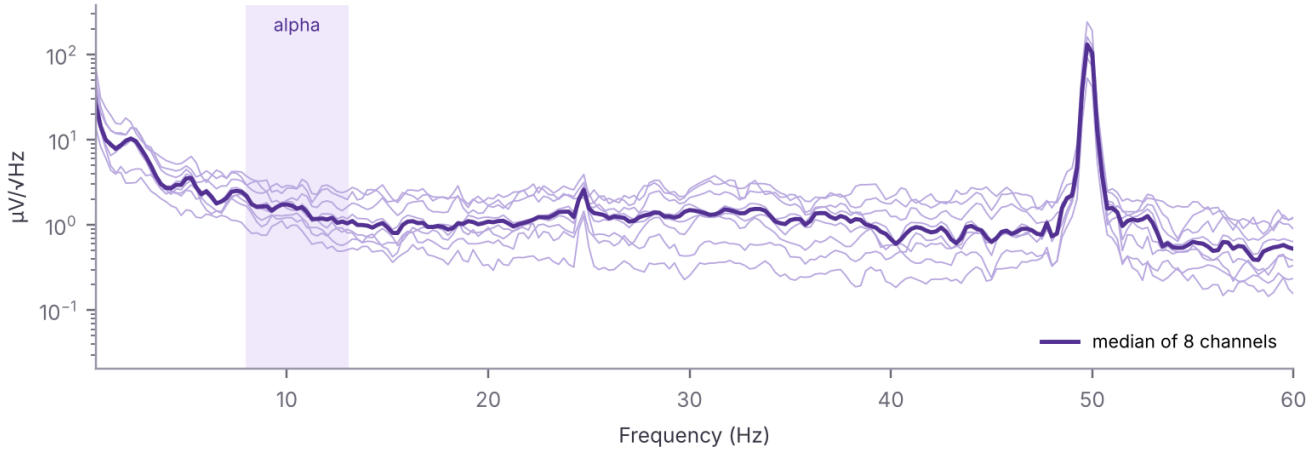


Figure 2 — Amplitude spectral density of output1.xlsx (Welch, 4 s segments); bold = median of 8 channels. The 50 Hz peak is mains interference in that recording — use a notch filter.

09 RELATED DOCUMENTS

JN-DS-001 Product Datasheet	**JN-MD-002** Mechanical & Dimensional Drawing	**JN-PCB-003** PCB & Manufacturing Specification
JN-PIN-004 Connector Pinout & Electrode Guide	**JN-QSG-005** Quick Start & Safety Guide	

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